Current Results of Proton Beam Irradiation of Uveal Melanomas

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Abstract: Proton beam irradiation has been used for the treatment of 241 uveal melanomas over the past 7½ years. Twelve melanomas (5%) were small, 99 (41%) medium, 103 (43%) large and 27 (1%) extra-large melanomas. The mean length of follow-up was 21 months and the median 15 months. Ninety-four percent of the treated lesions with a follow-up more than two years and 65% of tumors with shorter follow-up showed regression. The most recent visual acuity was 20/40 or better in 47% and 20/100 or better in 66%. Ten eyes were enucleated because of complications (9) or continued tumor growth (1). Thirteen patients developed metastases from 4 to 50 months of treatment. Our data indicate that proton irradiation can be used to treat melanomas of various sizes and in a variety of locations, and preliminary results suggest that proton therapy has no deleterious effect on the likelihood of the development of metastases. [Key words: irradiation, melanoma, proton beam, treatment, uvea.] Ophthalmology 92:284–291, 1985

Enucleation has long been the usual treatment for posterior malignant uveal melanoma. The advantage of this method in terms of increased life expectancy has been recently challenged[1] and alternative forms of treatment have been used with increased frequency during the past few years.[2–8] Photocoagulation[2] and local resection[3] have been used in selected instances, but radiotherapy remains the most widely used method.[4–8]

There are two major radiotherapeutic techniques for the treatment of uveal melanomas. Radioactive plaques can be sutured on the sclera over the area of the tumor[4–6] and particulate radiations such as protons[8] and helium ions[7] can be used. The advantages of the latter modalities are premised on better dose distributions between tumor and normal tissues. The dose delivered with protons or helium ions can be localized to the tumor more accurately, excluding from the radiation field all or some of the radiosensitive intraocular structures.[7,8] Hopefully, this would improve the therapeutic ratio of local control versus complications.

Proton beam irradiation has been used for the treatment of uveal melanomas at the Harvard Cyclotron since 1975, and its application has increased in frequency during the past few years.[8–11]

We report here the current results of treatment of 240 patients with uveal melanoma (241 eyes).

MATERIALS AND METHODS

All patients were treated between July 1975 and December 1982. Follow-up data were analyzed up to June 1983. The 122 men and 118 women treated ranged in age from 14 to 83 years (mean and median age, 56 years). Most of the patients were in the sixth and seventh decades of life. All patients were white. The right eye

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